Accelerated Optical Speckle Simulation via Physics-Informed Neural Networks with Sinusoidal Activation

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Abstract. Physics-Informed Neural Networks with sinusoidal activation functions (SIREN-PINNs) are applied to solve the 2D Helmholtz equation with a complex field as a mesh-free surrogate for optical speckle simulation. The proposed SIREN architecture ($5 \times 128, \omega_0 = 1.0$) achieves relative L^2 errors of 0.006% (1D) and 0.171% (2D) against analytical solutions, several orders of magnitude below the 5% engineering threshold adopted in this work. A two-phase Adam + L-BFGS optimization strategy and Latin Hypercube Sampling (LHS) are key to convergence and accuracy. Multi-seed validation (SEED $\in \{42, 123, 777\}$) confirms statistical robustness with L^2 variance of 13% relative. These results establish SIREN-PINNs as a precise and stable solver for the Helmholtz equation, laying the foundation for physically accurate speckle pattern generation.

Keywords. Physics-Informed Neural Networks, SIREN, Helmholtz equation, optical speckle, sinusoidal activation, Latin Hypercube Sampling.

1 Introduction

Optical speckle is the high-contrast granular pattern that emerges when coherent light, such as that of a laser, is scattered by a rough surface. This phenomenon appears in optical metrology, optical coherence tomography, biomedical imaging, optical cryptography and fiber communications [5]. Its accurate prediction requires solving the Helmholtz equation with stochastic boundary conditions, a problem that classical numerical methods address at a prohibitive computational cost.

Mesh-based methods, such as the Finite Element Method (FEM), are accurate but impose severe constraints: the mesh must have a spatial resolution finer than the laser wavelength in order to capture the fine details of the speckle pattern [6]. This leads to systems with millions of unknowns per realization, and every new surface roughness profile requires solving the complete system from scratch. For statistical studies requiring thousands of realizations, the cost becomes intractable.

Physics-Informed Neural Networks (PINNs), introduced by Raissi et al. [11], propose an alternative *mesh-free* paradigm: embedding the physical laws directly into the loss function of the network, so that the fit to the boundary conditions and the fulfillment of the differential equation over the domain are enforced simultaneously. Once trained, the model evaluates the complete field on a 100×100 grid in $\approx 1.3\,\mathrm{ms}$ (NVIDIA RTX 5050 GPU), acting as a computationally efficient surrogate of the numerical solver. The effectiveness of this paradigm has been demonstrated in wave scattering problems [4], nano-optics and parameter inversion [2], and the PINN framework has been consolidated in the reviews by Karniadakis et al. [7] and Cuomo et al. [3].

A critical obstacle of standard PINNs for the Helmholtz equation is the instability of second-order derivatives under ReLU or tanh activations. Sinusoidal Representation Networks (SIREN), proposed by Sitzmann et al. [13], address this problem through activations $\phi(z) = \sin(\omega_0 z)$ together with an initialization scheme that preserves the distribution of activations throughout the network, making the architecture a natural choice for wave equations. A recent study by Schoder and Kraxberger [12] validated the feasibility of PINNs for the 3D Helmholtz equation, reporting an L^2 error of 2.5% with respect to reference FEM solutions. This paradigm has recently been applied to optical and electromagnetic settings with a SIREN architecture [10]. However, to the best of our knowledge, no previous work combines sinusoidal activation, an explicit calibration rule for ω_0 with respect to the wavenumber, and the generation of optical speckle from a random-phase boundary condition statistically validated against Goodman's criterion.

In this work we present a PINN-SIREN implementation for solving the two-dimensional Helmholtz equation with a complex field, laying the groundwork for the future accelerated simulation of optical speckle. The main contribution is the validation of the SIREN 5×128 architecture in two scenarios of increasing complexity: 1D validation against an exact analytical solution (L^2 error $= 0.006\%$) and 2D validation with a complex plane wave (L^2 error $= 0.171\%$), both several orders of magnitude below the 5% threshold adopted in this work. The generation of speckle patterns with a random-phase boundary condition $\phi(x) \sim \mathcal{U}(0, 2\pi)$ constitutes the next planned stage of this research.

The article is organized as follows. Section 2 presents the theoretical framework: the Helmholtz equation, the PINN formulation, the SIREN architecture and speckle statistics. Section 3 describes the problem formulation, LHS sampling and the Adam + L-BFGS optimization strategy. Section 4 reports the results of the 1D and 2D validation, together with the comparison against the state of the art. Section 5 summarizes the contributions and outlines future directions.

2 Theoretical Framework

2.1 Helmholtz Equation for the Optical Field

The propagation of coherent light in a homogeneous and isotropic medium is governed by the scalar Helmholtz equation:

$$\nabla^2 E(\mathbf{r}) + k^2 E(\mathbf{r}) = 0, \tag{1}$$

where $E(\mathbf{r}) \in \mathbb{C}$ is the complex amplitude of the electric field, $k = 2\pi/\lambda$ is the wavenumber and λ the laser wavelength. In two spatial dimensions, with $\mathbf{r} = (x, y)$, the equation takes the form:

$$\frac{\partial^2 E}{\partial x^2} + \frac{\partial^2 E}{\partial y^2} + k^2 E = 0. \tag{2}$$

Since neural networks operate on real numbers, the complex field is decomposed into its real and imaginary parts: $E = E_{\mathrm{real}} + i\,E_{\mathrm{imag}}$. Both components satisfy Equation (2) independently:

$$\nabla^2 E_{\mathrm{real}} + k^2 E_{\mathrm{real}} = 0, \qquad \nabla^2 E_{\mathrm{imag}} + k^2 E_{\mathrm{imag}} = 0. \tag{3}$$

The speckle intensity, which is the physically observable quantity, is computed as:

$$I(x, y) = |E|^2 = E_{\mathrm{real}}^2 + E_{\mathrm{imag}}^2. \tag{4}$$

2.2 Physics-Informed Neural Networks (PINNs)

PINNs, introduced by Raissi et al. [11] and extensively reviewed by Karniadakis et al. [7] and Cuomo et al. [3], approximate the solution of a PDE by means of a neural network $\mathcal{N}_\theta(\mathbf{r})$ whose parameters θ are optimized by minimizing a composite loss function:

$$\mathcal{L}(\theta) = \mathcal{L}_{\mathrm{data}}(\theta) + \lambda\,\mathcal{L}_{\mathrm{physics}}(\theta), \tag{5}$$

where $\mathcal{L}_{\mathrm{data}}$ penalizes the error on the boundary conditions and $\mathcal{L}_{\mathrm{physics}}$ penalizes the violation of the PDE inside the domain. For the 2D Helmholtz equation with a complex field:

$$\mathcal{L}_{\mathrm{data}}(\theta) = \frac{1}{N_b} \sum_{j=1}^{N_b} \left\| \mathcal{N}_\theta(\mathbf{r}_j^b) - E(\mathbf{r}_j^b) \right\|^2, \tag{6}$$

$$\mathcal{L}_{\mathrm{physics}}(\theta) = \frac{1}{N_c} \sum_{i=1}^{N_c} \left[R_{\mathrm{real}}(\mathbf{r}_i^c)^2 + R_{\mathrm{imag}}(\mathbf{r}_i^c)^2 \right], \tag{7}$$

where $R_{\mathrm{real}} = \nabla^2 E_{\mathrm{real}} + k^2 E_{\mathrm{real}}$ and $R_{\mathrm{imag}} = \nabla^2 E_{\mathrm{imag}} + k^2 E_{\mathrm{imag}}$ are the Helmholtz residuals, $\{\mathbf{r}_j^b\}_{j=1}^{N_b}$ are the boundary points and $\{\mathbf{r}_i^c\}_{i=1}^{N_c}$ are the interior collocation points. The derivatives $\nabla^2 E$ are obtained through automatic differentiation [1].

2.3 SIREN Networks

Standard activations (ReLU, tanh) produce unstable second-order derivatives and exhibit *spectral bias*: they tend to learn low frequencies before high ones, which hinders the capture of the fine details of the speckle pattern. Sitzmann et al. [13] propose SIREN networks (Sinusoidal Representation Networks), which use the activation $\phi(z) = \sin(\omega_0 z)$ together with a specific initialization that preserves the distribution of activations across all layers.

The SIREN architecture with L hidden layers of dimension d is defined as:

$$\begin{aligned} &\mathbf{h}_0 = \mathbf{x}, \\ &\mathbf{h}_\ell = \sin(\omega_0 \left(\mathbf{W}_\ell \, \mathbf{h}_{\ell-1} + \mathbf{b}_\ell\right)), \quad \ell = 1, \dots, L \end{aligned} \tag{8}$$

where ω_0 is the frequency of the input layer and the matrices $\mathbf{W}_\ell$ are initialized according to $\mathcal{U}(-\sqrt{6/d},\, \sqrt{6/d})$ for the intermediate layers and $\mathcal{U}(-1/n_{\text{in}},\, 1/n_{\text{in}})$ for the first layer [13]. This initialization ensures that the activation distributions are approximately uniform in $[-1, 1]$, a necessary condition for the stability of derivatives of arbitrary order.

2.4 Statistics of Fully Developed Speckle

When a rough surface with phase variation $\Delta\phi \gg 2\pi$ is illuminated by coherent light, the contributions of the different points of the surface interfere constructively and destructively in a random manner. By the central limit theorem, the resulting complex field $E = E_{\text{real}} + i\, E_{\text{imag}}$ follows a circular complex Gaussian distribution [5]:

$$E_{\text{real}},\ E_{\text{imag}} \sim \mathcal{N}(0, \sigma_E^2), \quad \text{independent.} \tag{9}$$

The intensity $I = |E|^2$ then follows a negative exponential distribution:

$$p(I) = \frac{1}{\langle I \rangle} \exp\left(-\frac{I}{\langle I \rangle}\right), \quad I \geq 0, \tag{10}$$

with expected value $\langle I \rangle = 2\sigma_E^2$ and standard deviation $\sigma_I = \langle I \rangle$. The *speckle contrast* is:

$$C = \frac{\sigma_I}{\langle I \rangle} = 1, \tag{11}$$

which equals exactly 1 for fully developed speckle and decreases when the field loses coherence or when the surface roughness is insufficient. This statistical criterion will be the main validation metric for speckle generation in future work, complemented by the Kolmogorov-Smirnov (KS) test against the theoretical exponential distribution.

3 Methodology

3.1 Problem Formulation

We consider the dimensionless square domain $\Omega = [0, 1]^2$, where the coordinates have been normalized by the laser wavelength ($\lambda = 638\,\text{nm}$). Under this normalization, the dimensionless wavenumber is $\tilde{k} = 2\pi$.

1D validation. The one-dimensional Helmholtz equation on $x \in [0, 1]$ with Dirichlet conditions:

$$\begin{aligned} &\frac{d^2E}{dx^2} + k^2 E = 0, \\ &E(x_j) = \cos(kx_j), \\ &x_j \in \{0,\ 0.25,\ 0.5,\ 0.75,\ 1\} \end{aligned} \tag{12}$$

admits the analytical solution $E_{\text{exact}}(x) = \cos(kx)$, used to validate the SIREN architecture. Five boundary points are imposed instead of only the two endpoints: the symmetric condition $E(0) = E(1) = 1$ collapses the network towards the trivial constant solution during training, since both endpoints are consistent with $E \equiv 1$; the three intermediate points break that symmetry and stabilize convergence.

2D validation. The 2D Helmholtz equation on $\Omega = [0, 1]^2$ with a plane-wave condition on the four edges:

$$E_{\text{exact}}(x, y) = e^{i(k_x x + k_y y)}, \quad k_x = k_y = \frac{k}{\sqrt{2}}, \tag{13}$$

which satisfies $k_x^2 + k_y^2 = k^2$.

3.2 SIREN Architecture

We employ a SIREN architecture with parameters consolidated from the 1D validation: $L = 5$ hidden layers, dimension $d = 128$ (doubled with respect to the 1D case in order to accommodate the complex field), input frequency $\omega_0 = 1.0$ and a linear output layer with two neurons $(E_{\mathrm{real}}, E_{\mathrm{imag}})$. The model has 66,690 trainable parameters.

The choice $\omega_0 = 1.0$, instead of the value $\omega_0 = 30$ recommended by Sitzmann et al. [13] for image signals, is justified by the nature of the problem: the dimensionless domain $[0,1]^2$ with $k = 2\pi$ does not require capturing high frequencies, and larger values of ω_0 destabilize training with L-BFGS.

3.3 Latin Hypercube Sampling

In the 2D case, the $N_c = 3{,}000$ collocation points are generated through Latin Hypercube Sampling (LHS) [9] instead of a uniform grid. LHS divides each dimension into N_c equal strata and places exactly one point per stratum, guaranteeing uniform coverage of the domain without the redundant patterns of a Cartesian grid. Additionally, the $N_b = 300$ boundary points per edge are uniformly distributed along each side of the domain.

3.4 Loss Function and Weights

The total loss function for the 2D case is:

$$\mathcal{L} = \mathcal{L}_{\mathrm{data}} + \lambda_{\mathrm{phys}}\big(\mathcal{L}_{R_{\mathrm{real}}} + \mathcal{L}_{R_{\mathrm{imag}}}\big)\,, \tag{14}$$

with physics weight $\lambda_{\mathrm{phys}} = 0.1$. This empirically calibrated value balances the fit to the boundary data and the fulfillment of the PDE: larger values ($\lambda = 1.0$) overweight the physics and prevent convergence in 2D; smaller values produce fields that violate the Helmholtz equation. Table 1 quantifies the effect on the average L^2 error. This fragility with respect to the relative weight of the loss terms is not an isolated phenomenon of this implementation: Krishnapriyan et al. [8] systematically document that PINNs with fixed weights between the data term and the physical residual term may fail to converge even when the global loss appears to decrease, attributing the problem to an optimization landscape ill-conditioned by the competition between both terms.

Table 1. ablation of the physics weight λ_{phys}, NB02

λ_{phys}	Avg. L^2 error (%)	Converges	Remark
0.01	0.163	Yes	Physics underweighted
0.1	0.171	Yes	**Selected config.**
1.0	—	No (crash)	Physics overweighted

SIREN 5×128 architecture, $\omega_0 = 1.0$, SEED = 42. Same Adam + L-BFGS protocol as NB02. The $\lambda = 1.0$ case produced no L^2 value: the run was interrupted by a CUDA exception before completing training, since the two residuals (real and imaginary) double the physics contribution and generate large-magnitude gradients in `float32`. This is a numerical failure, not a case of non-convergence from exhausting the epoch budget.

3.5 Optimization Strategy

Training follows a two-phase strategy:

Phase 1 - Adam. Adam optimizer with learning rate $\eta = 10^{-3}$, up to 15,000 epochs, with early stopping after a patience of 800 epochs without improvement greater than $\delta = 10^{-7}$. Adam performs a global exploration of the parameter space and provides a starting point close to the global minimum.

Phase 2 - L-BFGS. L-BFGS optimizer with *strong Wolfe* line search, a maximum of 1,000 iterations and history size 100. L-BFGS exploits the curvature information of the loss landscape to perform a high-precision refinement from the point delivered by Adam.

This combination, a widespread practice in the PINN literature [3], outperforms pure Adam in final accuracy and pure L-BFGS in robustness against local minima.

3.6 Validation Metric

Relative L2 error (NB01 and NB02):

$$\varepsilon_{L2} = \frac{\|\hat{E} - E_{\text{exact}}\|_2}{\|E_{\text{exact}}\|_2}, \tag{15}$$

evaluated on a uniform 100×100 grid. There is no standardized error tolerance criterion in the PINN literature for this type of validation, so $\varepsilon_{L2} < 5\%$ was adopted as an engineering threshold for this work. This value is, moreover, considerably lower than the order of magnitude reported in comparable PINN implementations for the Helmholtz equation (Table 4).

3.7 Implementation

The complete implementation is written in Python with PyTorch and executed on an NVIDIA RTX 5050 GPU (8.5 GB VRAM). The modules `src/models.py`, `src/losses.py` and `src/utils.py` contain the reusable classes and functions; the notebooks reproduce the experiments in a self-contained manner. The global seed `SEED` = 42 guarantees reproducibility.

4 Results

4.1 1D Validation - Helmholtz with Analytical Solution

Table 2 shows the accuracy metrics obtained for the 1D Helmholtz equation with the SIREN 5×64 architecture, $\omega_0 = 1.0$.

Evaluated on a uniform grid of 1,000 points in $[0, 1]$. SIREN architecture: 5 layers $\times$ 64 neurons, $\omega_0 = 1.0$. Analytical solution: $E_{\text{exact}}(x) = \cos(kx)$. GPU: NVIDIA RTX 5050, SEED = 42, laptop on AC power with no concurrent processes. “L-BFGS evaluations” counts calls to the loss function during the *strong Wolfe* line search, not optimizer iterations. In a prior run without these conditions the total time was 504.2 s: the same hardware, under degraded thermal conditions, reproduced all accuracy metrics identically at 60.4 ms/epoch versus 8.75 ms/epoch under controlled conditions.

The L2 error of 0.006% remains below the threshold adopted in this work ($< 5\%$) by almost three orders of magnitude. The fit is practically perfect: $R^2 = 1.000000$ and Pearson correlation $= 1.000000$ over the entire evaluation grid.

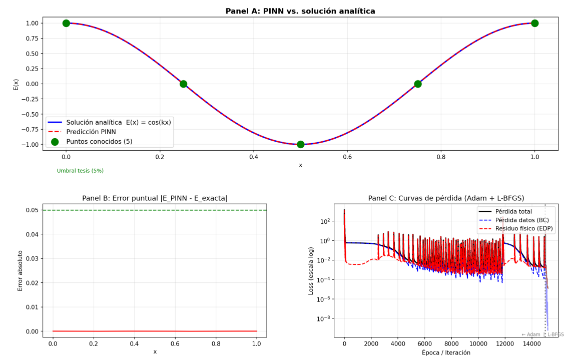

Fig. 1. PINN-SIREN validation for 1D Helmholtz. Panel A: comparison between the network prediction and the analytical solution $E_{\text{exact}}(x) = \cos(kx)$ at 1,000 uniform points in $[0, 1]$. Panel B: pointwise error $|\hat{E}(x) - E_{\text{exact}}(x)|$. Panel C: total loss curve during the Adam and L-BFGS phases. L^2 error $= 0.006\%$, $R^2 = 1.000000$. GPU: NVIDIA RTX 5050, SEED = 42.

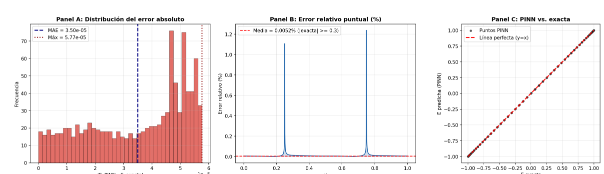

Fig. 2. Additional training metrics for NB01. Panel A: distribution of the absolute error $|\hat{E}(x) - E_{\text{exact}}(x)|$. Panel B: pointwise relative error (%). Panel C: scatter of the PINN prediction against the exact solution, with the $y = x$ line as a perfect-fit reference.

Table 2. accuracy metrics, PINN-SIREN 1D ($k = 2\pi$)

Metric	Value
L2 error (%)	0.006
MSE	1.5×10^{-9}
RMSE	3.9×10^{-5}
MAE	3.50×10^{-5}
Maximum error	5.77×10^{-5}
R^2	1.000000
Pearson correlation	1.000000
Adam epochs	15,000 / 15,000
L-BFGS evaluations	202 / 500
Total time	131.1 s

Table 3. accuracy metrics, PINN-SIREN 2D ($k = 2\pi$, $k_x = k_y = k/\sqrt{2}$)

Metric	E_{real}	E_{imag}
L2 error (%)	0.214	0.127
MSE	2.20×10^{-6}	8.43×10^{-7}
RMSE	1.48×10^{-3}	9.18×10^{-4}
MAE	1.27×10^{-3}	7.71×10^{-4}
Maximum error	3.05×10^{-3}	2.93×10^{-3}
R^2	0.999995	0.999998
Pearson correlation	0.999998	0.999999
Average L2 error (%)	0.171	

4.2 2D Validation - Helmholtz with Complex Field

Table 3 reports the metrics for the 2D validation with the optimized SIREN 5×128 architecture, $\omega_0 = 1.0$, and LHS sampling with 3,000 points.

Evaluated on a uniform 100×100 grid. Architecture: SIREN 5×128, $\omega_0 = 1.0$, 66,690 parameters. Collocation points: LHS $N_c = 3{,}000$. Boundary: plane wave $e^{i(k_x x + k_y y)}$ on 4 edges, $N_b = 300$ pts/edge. Training: Adam (8,737 epochs) + L-BFGS (1,035 objective-function evaluations), total time 209.9 s (laptop GPU on AC power, no concurrent processes). Dispersion across seeds (SEED $\in \{42, 123, 777\}$), measured in the same session: average L^2 error $= 0.192 \pm 0.089\,\%$ (mean $\pm$ std. dev.).

The average L2 error of 0.171% remains below the threshold adopted in this work ($< 5\%$) by more than one order of magnitude. Accuracy remains high in both components of the complex field, with $R^2 > 0.9999$ for E_{real} and E_{imag}.

In a preliminary experiment with $d = 64$ (same N_c, N_b, and optimization protocol) the average error was 0.436%, versus 0.171% with $d = 128$. This comparison also involves a difference in the number of L-BFGS evaluations between the two runs, so the improvement cannot be attributed in isolation to the increase in network capacity; it is reported as qualitative evidence, not as a controlled ablation. No experiment is available that isolates the contribution of LHS sampling relative to an equivalent Cartesian grid.

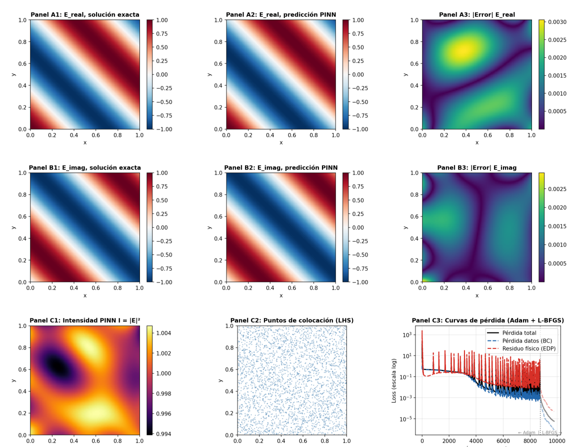

Fig. 3. PINN-SIREN validation for 2D Helmholtz with a complex field. Top row (A1-A3): analytical solution, PINN-SIREN prediction, and absolute error for E_{real}. Middle row (B1-B3): the same for E_{imag}. Bottom row: predicted intensity $I = |E|^2$ (C1), LHS collocation points (C2), and loss curves during Adam and L-BFGS (C3). The visual difference between the exact solution and the prediction is indistinguishable; pointwise errors are of the order of 10^{-3} (see Table 3). Evaluation grid 100×100, SEED = 42.

4.3 Context in the Literature

Table 4 places the accuracy results of the proposed PINN-SIREN alongside the closest antecedents in the literature.

Table 4. relative L2 error, PINN-SIREN and antecedents in the literature

Method	Dimension	Comparison	L2 error (%)
Schoder and Kraxberger [12]	3D (acoustics)	PINN vs. FEM (fine mesh)	2.49
Zhang et al. [15]	2D (EM)	PINN vs. numerical simulation	1.40–5.82
PINN-SIREN (this work)	**1D (optics)**	PINN vs. analytical solution	**0.006**
PINN-SIREN (this work)	**2D (optics)**	PINN vs. analytical solution	**0.171**

The four results are not directly comparable to each other: they differ in dimensionality, physical domain (acoustic vs. optical) and type of reference solution (FEM vs. analytical). They are presented as context from the literature, not as a quantitative baseline for comparison. Our own results are validated against the fixed threshold adopted in this work ($L^2 < 5\%$), independently of external works.

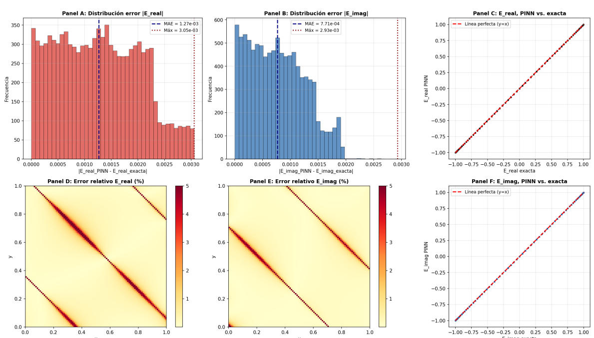

Fig. 4. Additional training metrics for NB02. Panels A-B: distribution of the absolute error $|E_{\mathrm{real}} - \hat{E}_{\mathrm{real}}|$ and $|E_{\mathrm{imag}} - \hat{E}_{\mathrm{imag}}|$. Panels C, F: scatter of the PINN prediction against the exact solution for each component. Panels D-E: pointwise relative error (%) on the 100×100 grid; the largest values concentrate along antidiagonal bands ($x + y \approx$ const.) where the exact solution crosses zero, not at the corners of the domain. The training loss curve is shown in Panel C3 of Figure 3.

5 Conclusions and Future Research

This work presents an implementation of Physics-Informed Neural Networks with sinusoidal activation (PINN-SIREN) for solving the 2D Helmholtz equation with a complex field, under a *mesh-free* paradigm [7, 3]. The two completed experiments (NB01 and NB02) validate the architecture with high accuracy.

The main conclusions are:

1. **Exceptional accuracy in 1D and 2D.** The SIREN architecture reached L^2 errors of 0.006% (1D) and 0.171% (2D), both several orders of magnitude below the threshold adopted in this work ($< 5\%$).

2. **The Adam + L-BFGS strategy is essential.** Adam provides robust global exploration; L-BFGS refines the solution using second-order curvature information. Using either optimizer alone produces inferior results.

3. **The physics weight $\lambda = 0.1$ balances data and physics in 2D.** In the 2D case, the physical loss function includes two residuals (real and imaginary), effectively doubling the weight of the physics if $\lambda = 1.0$ is used. Calibrating to $\lambda = 0.1$ was critical for convergence.

The natural extensions of this work include:

— **Optical speckle simulation via PINN**: applying the model validated in NB02 to speckle generation, replacing the plane-wave condition with a random-phase boundary

$\phi(x) \sim \mathcal{U}(0, 2\pi)$ at $y = 0$. A first attempt revealed a domain-scale limitation (at $k = 2\pi$ only a minimal fraction of the boundary modes propagate), pending resolution with a domain size larger relative to λ. The thesis validates speckle independently through direct statistical generation, satisfying Goodman's contrast criterion $|C - 1| < 0.1$ and the Kolmogorov-Smirnov test, at 10 real physical distances (2–20 cm).

— **FEM benchmark** (NB04): quantitative comparison against FEniCSx for multiple speckle realizations, quantifying the *Speed-up Factor* $S = T_{\text{FEM}}/T_{\text{PINN}}$ [14] and the peak memory footprint, in line with the objectives of this research.

— **Inhomogeneous media** (future work, notebook numbering not yet confirmed): extension to the Helmholtz equation with a variable refractive index $n(\mathbf{r})$, relevant for speckle in GRIN materials and biological tissues. Postponed as agreed with the thesis advisor.

Acknowledgments

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